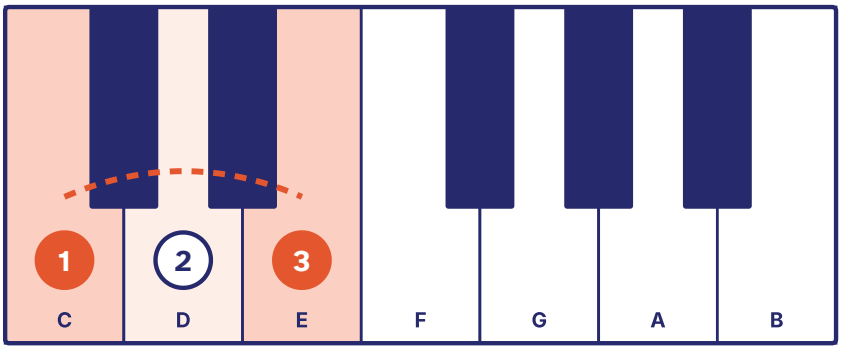
PART III · SCALES AND KEYS · 1

Counting intervals

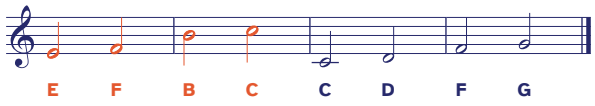
How far apart are two notes? That gap is an **interval**. The very next key is a **semitone**; two make a **tone**. To number it, count the letter names, both ends included.

THE SECRET

Count the letters, both ends included: C to E is a 3rd.



C to E: count the letters, 1 2 3, a 3rd

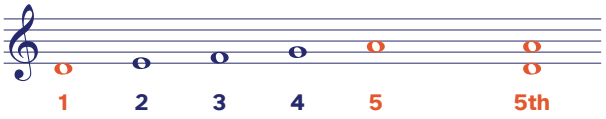


E–F and B–C are semitones; C–D and F–G are tones.



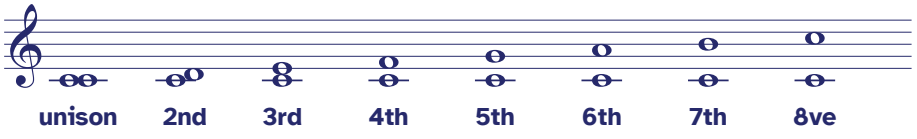
Melodic: one note after the other. Harmonic: together.

Counting the number



From **D** up to **A** (the red notes): D E F G A is five letters, so the interval is a 5th, whether the notes come one after the other or together.

Interval numbers from C

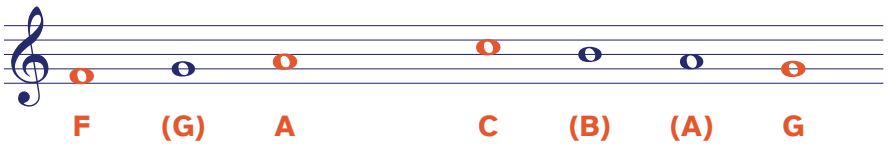


Your hand is a ruler: thumb on C, little finger on G: five fingers on five white keys, a 5th.

PART III · SCALES AND KEYS · 1 · CONTINUED

EXAMPLE 1

How far apart are these notes?

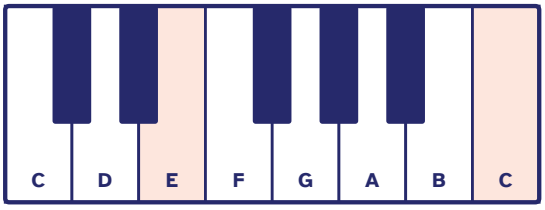


1. **Call the first note 1:** F is 1.
2. **Count every letter to the second note:** F G A is 3 letters: a 3rd.
3. **Going down works the same:** C B A G is 4 letters: a 4th.

F up to A is a **3rd**; C down to G is a **4th**. Sharps and flats never change the number.

EXAMPLE 2

Name an interval on the keyboard

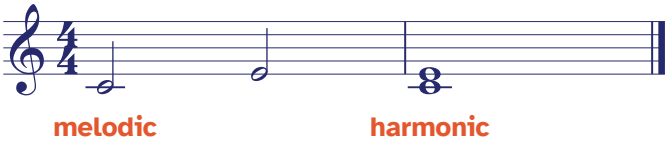


1. **Find both notes:** E and the C above it.
2. **Count the letters:** E F G A B C is six.
3. **Both ends count:** so the interval is a 6th.

E up to C is a **6th**.

EXAMPLE 3

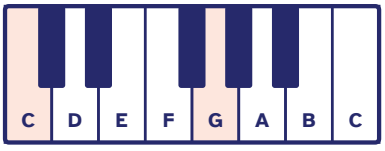
Melodic or harmonic?



1. **The first bar:** C then E, one after the other.
2. **The second bar:** C and E together.
3. **Count both:** C D E, a 3rd each time.

The same 3rd: **melodic** when played one after the other, **harmonic** when played together.

TRY IT · 10 SECONDS



Thumb on C, finger 5 on G.

Your hand is a ruler: thumb to little finger on white keys is a 5th. Play it, then thumb to finger 3: a 3rd.

PART III · SCALES AND KEYS · 1 · CONTINUED

GAME

Count the letters

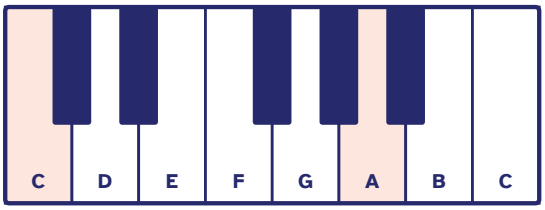
1	2	3	4
C up to G	D up to F	E up to the next E	G up to A

Count every letter, both ends included. What is each interval called?

Answer: 1 a 5th (C D E F G) · 2 a 3rd · 3 an octave (8) · 4 a 2nd.

EXAMPLE 4

Count an interval going down
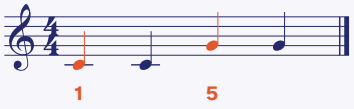


1. **Start on the top note:** A is 1.
2. **Count every letter down** to C: A G F E D C.
3. **Six letters:** a 6th.

A down to C is a **6th**, the same as C up to A.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The leap from the first “twinkle” to the second is C up to G: C D E F G, **five letters, a 5th**.

Try it. Sing the first two notes of any song, then count the letters between them.

THINK OF IT Monday to Wednesday is three days, not two: you count the day you start on. Intervals count the same way.

TRAP

- × Counting only the gaps, not the notes: always count both ends.
- × Letting a sharp or flat change the number: C to G♯ is still a 5th.

SEE ALSO

Major, minor and perfect p. 66 · Major scales p. 69

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Intervals

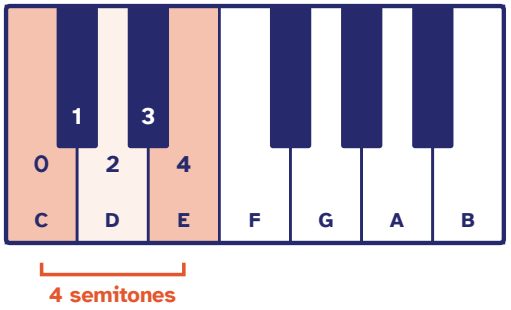
PART III · SCALES AND KEYS · 2

Major, minor and perfect

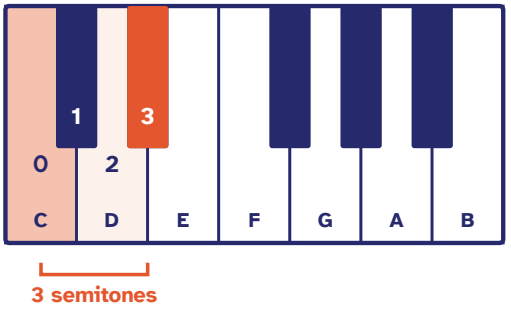
Two intervals with the same number can differ in size. C–E and C–E♭ are both 3rds, but one is a semitone wider. The **quality** (major, minor, perfect…) comes from counting **semitones**.

THE SECRET

Count the semitones: 4 make a major 3rd, 3 a minor 3rd.



From **C** to **E**: count every key, black and white. 4 semitones: a major 3rd.



From **C** to **E♭**: 3 semitones, one fewer: a minor 3rd.

One semitone less: a major 3rd becomes minor

Size in semitones

NUMBER	SMALLER	MAIN	WIDER	FIRST MET
unison		0 (perfect)		Level 1
2nd	1 (minor)	2 (major)		Level 1
3rd	3 (minor)	4 (major)		Level ½
4th		5 (perfect)	6 (augmented)	Level 1
5th	6 (diminished)	7 (perfect)		Level 1
6th	8 (minor)	9 (major)		Level 2
7th	10 (minor)	11 (major)		Level 2
octave		12 (perfect)		Level 1

2nds, 3rds, 6ths, 7ths
diminished ‹ **minor** ‹ **major** ‹ augmented

unisons, 4ths, 5ths, octaves
diminished ‹ **perfect** ‹ augmented

Each step on the ladder is one semitone. Six semitones (an augmented 4th or diminished 5th) is the tritone. Turned upside down, an interval's numbers add to 9 and major and minor swap: a major 3rd becomes a minor 6th.

PART III · SCALES AND KEYS · 2 · CONTINUED

EXAMPLE 1

Major or minor 3rd?



1. **Count the letters:** C D E is three, so it is a 3rd.
2. **Count the semitones:** C to C♯, D, E♭ is three.
3. **Three semitones** make a minor 3rd; four would be major.

C to E♭: a **minor 3rd**.

GAME

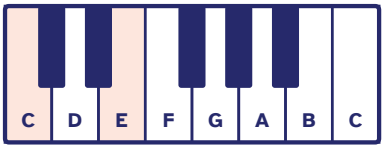
Perfect, major or minor?

1	2	3	4
C–G	D–F	C–A	E–A

Count the letters for the number, then the semitones for the size.

Answer: 1 perfect 5th (7) · 2 minor 3rd (3) · 3 major 6th (9) · 4 perfect 4th (5).

TRY IT · 10 SECONDS

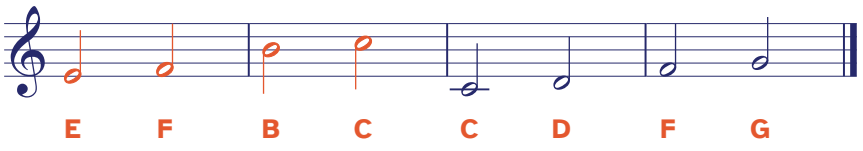


Play C–E, then C–E♭.

Bright, then dark. One key makes the difference: the major 3rd has four semitones, the minor 3rd three.

EXAMPLE 2

Tone or semitone?



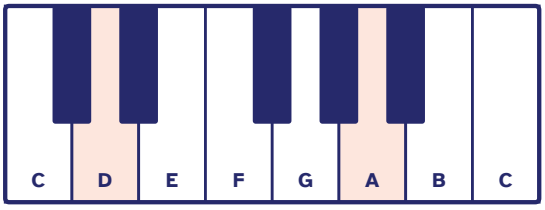
1. **E to F:** the very next key, no black key between: a semitone.
2. **C to D:** one key between (C♯): a tone.
3. **B to C** and **F to G**: check the same way.

E–F and B–C are semitones; C–D and F–G are tones.

PART III · SCALES AND KEYS · 2 · CONTINUED

EXAMPLE 3

Name D up to A

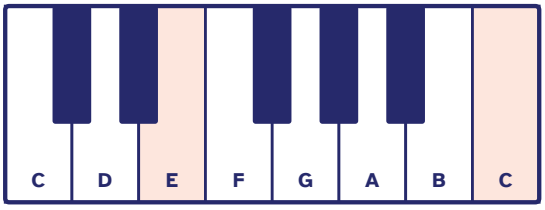


1. **The number:** D E F G A, five letters, a 5th.
2. **The size:** count every key from D to A, black and white: 7 semitones.
3. **7 semitones** is the main size of a 5th: perfect.

D up to A is a **perfect 5th**.

EXAMPLE 4

Name E up to C
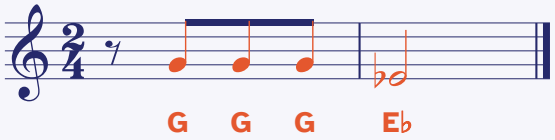


1. **The number:** E F G A B C, six letters, a 6th.
2. **The size:** 8 semitones from E to C.
3. **A major 6th** has 9, so 8 is one smaller: minor.

E up to C is a **minor 6th**.

YOU KNOW THIS

Beethoven, Symphony No. 5



Da-da-da-DUM: G down to E♭ is four semitones, a **major 3rd**. Three semitones would be minor.

Try it. Count the keys from G down to E♭: F♯, F, E, E♭.

THINK OF IT A major and a minor 3rd are a smile and a frown: the same face, one small change at the corners.

TRAP

- × Counting white keys only: count every key, black and white.
- × Calling a 4th or 5th “major”: they are perfect.

SEE ALSO

Counting intervals p. 63 · Chords in a key p. 92

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Intervals